



From Curiosity to Creation

VEX robotic team: Robotex Ninjas

Current team No.: V5RC 91276C

Previous team No.: VEX IQ 276C





Who we are

- Founded by *Charlie* and *Preston* in 2023
- Competed for three VEXIQ seasons
 - Full Volume 2023-2024, Judge's award @ U.S. Open
 - Rapid Relay 2024-2025, **Texas #1, World ranking #52**, Texas Regional Teamwork Champion, Skill Challenge Champion, Think Award
 - Mix & Match 2025-2026 **Texas #2, World ranking #97**, Texas Regional Teamwork Champion, Skill Challenge Champion, Think Award
- Three VEX World Championship appearances
- Two VEX World Championship Division Third Place
- Transitioning from VEX IQ to VEX V5RC in 2026-2027 Override season





More than just robotics



Local scrimmage event, 2026



Robotic summer camp, 2026

We compete, we connect, we teach and we serve





Outline

- Introduction to Robot and Robotics
- VEX Educational Robotics and Robot Competition
- Why Robotics and Robot Competition?
- Hands-on Workshop





Introduction to Robot and Robotics



Is this a robot?



VS.



Is this a robot?



VS.



Is this a robot?



VS.



Is this a robot?



VS.



Robot

- **Definition:**

- a device that automatically performs complicated, often repetitive tasks (as in an industrial assembly line) – *Merriam-Webster Dictionary*

- **Trivia**

- Where did the word “robot” first become famous?
 - a) A science-fiction movie
 - b) A Czech play**
 - c) A Japanese comic book
 - d) A newspaper article
- Who designed a mechanical knight more than 500 years ago?
 - a) Leonardo da Vinci**
 - b) Albert Einstein
 - c) Thomas Edison
 - d) Issac Newton

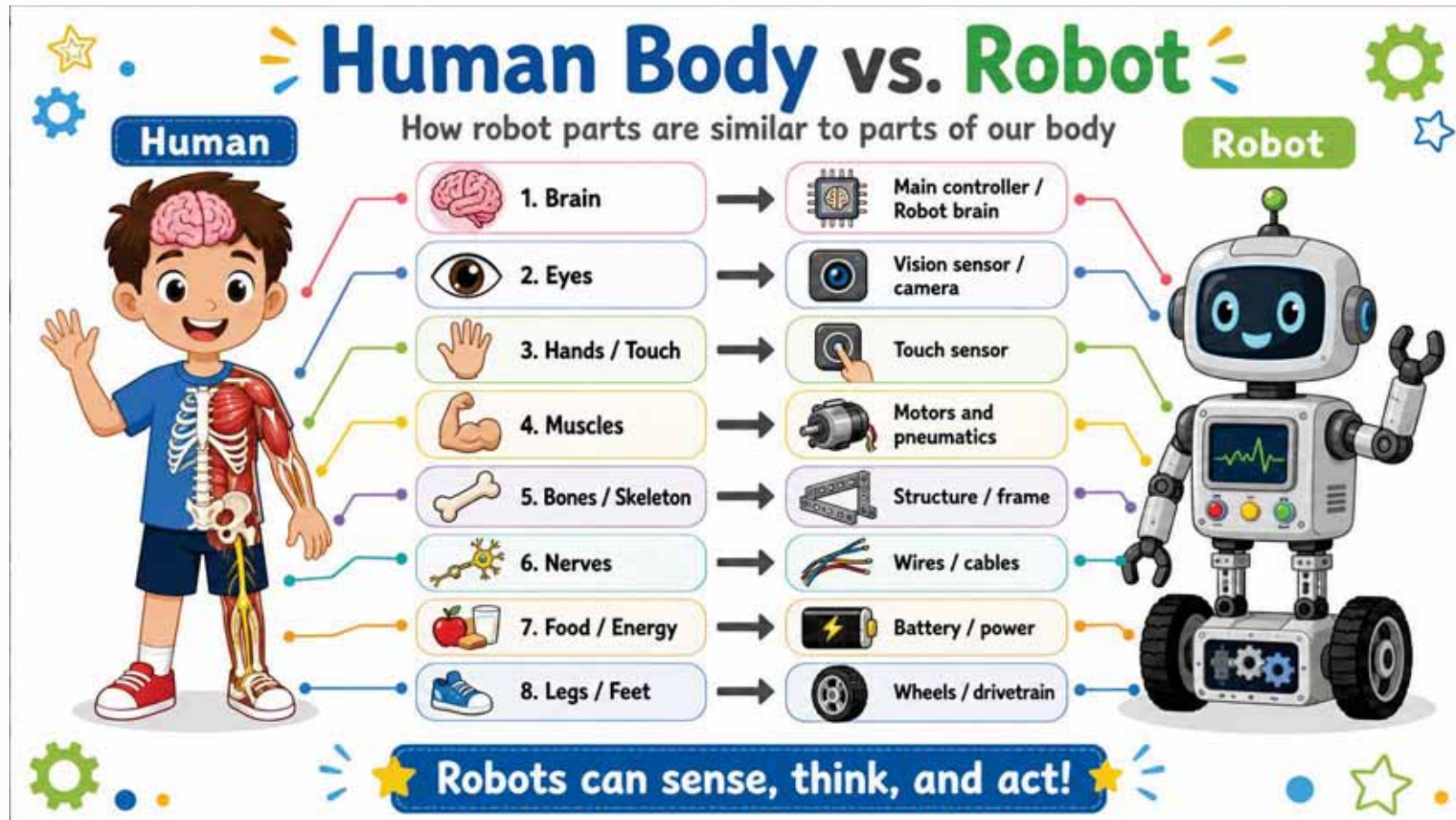


Machine vs. Robot

- Machine: a device with parts that work together to use energy and perform a task
- Robot: a special set of machine that can sense, process information and act in the physical world
- Robots are everywhere and fast growing



What a robot needs?





What Is Robotics?

Robotics is the study of how to build and program machines that can sense, think, and act.

Robotics is where we use:

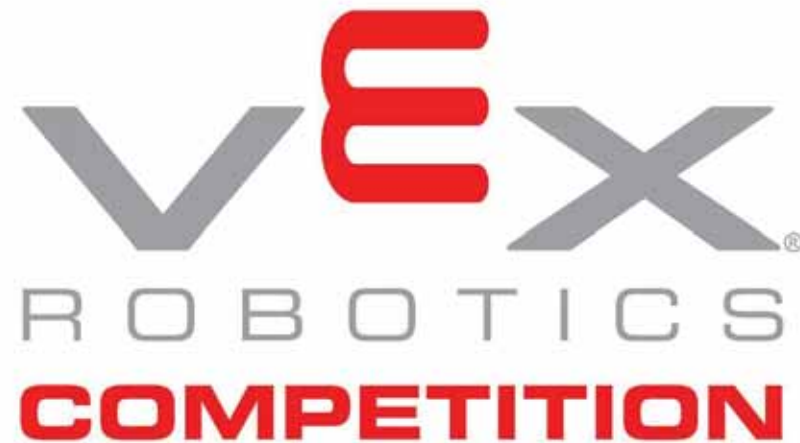
- **Mechanical design** — building the robot structure
- **Electrical parts** — motors, battery, brain, sensors
- **Coding** — giving instructions to the robot
- **Teamwork** — solving problems together

A robot is not just one thing. It is many ideas working together.





VEX Educational Robotics and Robot Competitions





What Is VEX Robotics?

VEX is a robotics education system for students. With VEX, students can:

- Build robots using real robot parts
- Add motors, wheels, gears, and sensors
- Code the robot to move and react
- Drive the robot in games and competitions
- Improve the robot through testing

VEX helps students learn engineering by actually building robots.

<https://www.vexrobotics.com/>



VEX robot competition





VEX competition robotics

- You can always find your group.
- New season, new game
- New rules and limitations

VEX 123 Grades Pre-K+	VEX GO Grades 3 - 5	VEX AIM Grades 4 - 12	VEX IQ Grades 5 - 8
 • Coding Starts Early • Pre-built Robot • Button & Card Coding • Block Coding	 • STEM Starts Early • Plastic Construction • Hands-on Activities • Blocks & Python	 • Real World Coding • Pre-built Robot • AI Vision • Blocks & Python	 • Applied STEM Learning • Plastic Construction • Worldwide Competitions • Blocks, Python, C++
VEX EXP Grades 8 - 12	VEX V5 Grades 9 - Uni	VEX CTE Grades 9 - Uni	VEX AIR Grades 9 - Uni
 • Real Classroom STEM • Metal Construction • Classroom Competitions • Blocks, Python, C++	 • Real World STEM • Worldwide Competitions • Metal Construction • Blocks, Python, C++	 • CTE & Career Readiness • Robotic Arm & Workcell • Sensors & Conveyors • Blocks, Python, C++	 • Object Manipulators • Video Controller • Dual Cameras • Blocks & Python

<https://www.vexrobotics.com/>





VEX Is Like a Robotics Toolbox



A VEX robot is made from many types of parts:

- **Structure parts** — the robot body
- **Motion parts** — wheels, gears, axles
- **Electronics** — brain, battery, motors
- **Sensors** — robot “eyes” and “touch”
- **Code** — instructions for the robot



VEX IQ vs. VEX V5

VEX IQ

- Plastic parts, no modification allowed
- Snap-together
- Good for younger students and beginners
- Great for learning robot building and coding



VEX V5

- Metal parts, freedom to modify parts, 3D printed parts and other customized parts are allowed
- Screws, nuts, shafts, motors, electronics
- More advanced
- Used by many middle school and high school teams



VEX IQ vs. VEX V5 *cont.*



VEX IQ

- More collaboration, consistency

VEX V5

- Dynamic strategy adjustment between defense and offense



Why Robotics and Robot Competition?



A robotic team needs many roles

- Builder
- Coder
- Driver
- Engineering notebooker
- CAD designer
- Outreach lead
- Scouter
- Pit crew





Why robotic competition?

- **Technical skills** (e.g., mechanical design, electronics, sensors, block and text coding, mathematics and measurement, testing and data collection)
- **Engineering design process** (brainstorm, design, build, test, identify problems, improve, test again, ..., document)
- **Teamwork skills** (communication, collaboration, leadership, time management, resolving disagreement, persistence after failure, responsibility, presentation)
- **Career Preparation** (problem-solving capabilities, project management, and technology literacy)
- **Fun** (Action-packed games, alliances partnership, “pit” atmosphere)





Hands-on Workshop

- Section #1: Code a robot
- Section #2: Build with VEX pieces
- Section #3: Drive a robot





Conclusions

- Curiosity is enough to begin
- Skills grow through practice
- You can form your own VEX robotic team

Acknowledgement:

- Little Elm Library and Miss. Diana Slavinsky
- Ninjas STEM Foundation
- You!



Questions?





Learn more about us!

